High Bandwidth Detection in Photon-Starved Environments Using Parametric Optical Amplification

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Research Experiences for Undergraduates

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Abstract

There is an increasing need to detect low level light signals at higher and higher speeds for applications ranging from biomedical imaging to underwater communications and sensing, and many others. Current technology, including avalanche photodiodes (PDs), photomultiplier tubes (PMTs), and amplified photodiodes cannot simultaneously provide low-photon-flux sensitivity and high bandwidth operation due to limitations in speed, sensitivity, or dynamic range. Here, we pair low-noise and high gain parametric amplification in optical fibers with detection using conventional, high-bandwidth photodiodes. We observe a **>10x** increase in sensitivity at bandwidths **>10x** larger than conventional technologies allowing us to resolve signals as low as **20 photons at 1.6 GHz bandwidth**. This method is drop-in compatible with any light detection system, making it ideal for integration with sensing, imaging, or telecommunications systems.

Current Technology

Applicable areas

1. Sensing and LIDAR in scattering media (underwater, UAVs)
2. Biomedical imaging
3. Low light level telecommunications

Technology	*Advantages*	*Disadvantages*
PMT	Single-photon resolution	Damage-limited bandwidth
Avalanche PD	Single-photon resolution	Cannot resolve photon number
Amplified PD	High bandwidth	Excess noise

Table 1: Current detection state of art and their pros and cons.

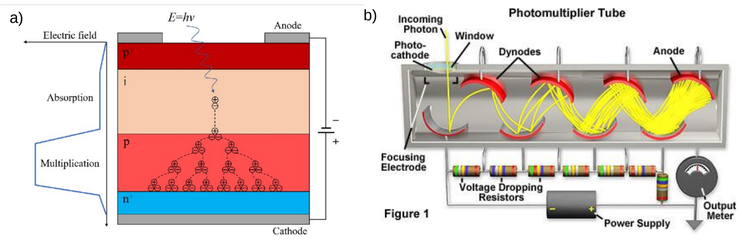


Figure 1 (a): Avalanche photodiode schematic (APD) [1]
(b): Photomultiplier tube schematic (PMT) [2]

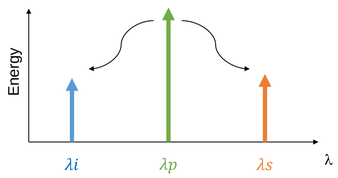
Approach: Parametric Amplification

Fiber Optical Amplification Methods:

- Erbium doped fiber amplifiers (EDFA)
- Raman amplification
- Parametric amplification with Four-Wave Mixing (FWM)

Four-wave mixing → high efficiency, low noise, wavelength agnostic

Energy Conservation:



$$\frac{2}{\lambda_p} = \frac{1}{\lambda_i} + \frac{1}{\lambda_s}$$

Figure 2: λp is the pump wavelength, λs is the Stokes wavelength, and λi is the Anti-Stokes wavelength.

Momentum Conservation:

$$\Delta\beta = 2\beta_p - \beta_s - \beta_i -= -2\gamma P_p \qquad \gamma = \frac{n_2\omega_p}{cA_{eff}}$$

β_s is the propagation constant of Stokes modes, β_i is the propagation constant of the Anti-Stokes mode, β_p is the propagation constant of the pump mode, γ is the nonlinear coefficient and P_p is the peak power of the pump laser. [3]

System Setup

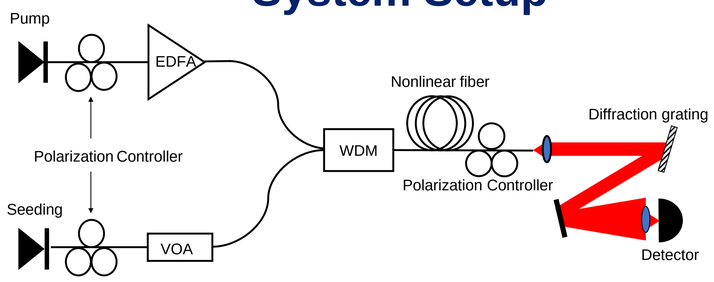


Figure 3: System architecture. Pump arm includes an erbium-doped fiber amplifier (EDFA) to generate high-peak-power pulses. It is combined with the seed using a wavelength division multiplexer (WDM). The seed uses a variable optical attenuator to achieve different input light levels. FWM occurs in a segment of highly nonlinear fiber. A diffraction grating removes all light outside of the anti-Stokes line at 1480 nm.

Specifications:

Pump source	
Wavelength (nm)	1545
Pulse Duration (ns)	0.5, 1, 2, 4, 8, 16
Frequency (MHz)	0.625-180

Seed source	
Wavelength (nm)	1620
Attenuation (dB)	0 - 60
Nominal power (μW)	20

Nonlinear fiber [4]	
Length (m)	50, 480
A_{eff} (μm^2)	12.4
λ_{ZDW} (nm)	1545

Detector	
Bandwidth (GHz)	5
Pass band (nm)	0.7
λ_c (nm)	1480

Table 2: A_{eff} is the effective area of the fiber. λ_{ZDW} is the zero dispersion wavelength of the fiber. λ_c is the central wavelength of the bandpass filter.

Determining NEP

Noise Equivalent Power (NEP) is the power for which signal to noise ratio (SNR) is unity, representing the minimum power that can be detected.

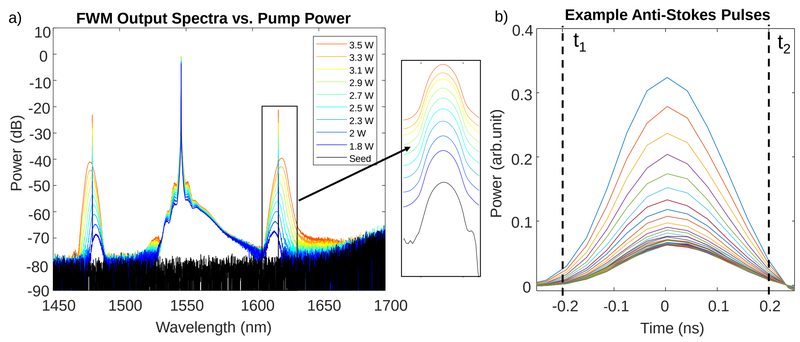


Figure 4 (a): FWM spectra for the system with a range of 1.8W~3.5W input pump peak power. Zoom-in of the Stokes line shown beside and we measure the gain by the difference between each peak pump power with the peak seed power. (b): Example output pulses for 0.5 ns duration as a function of seed power.

1. Measure anti-Stokes energy as a function of seed power, 100 times.
2. Calculate SNR from energy statistics → NEP occurs for SNR = 1.

$$E = \int_{t_1}^{t_2} dt\, P(t) \qquad SNR = \frac{\bar{E} - \overline{E_{no\ seed}}}{\sigma E_{no\ seed}} \qquad BW = \frac{1}{2(t_2 - t_1)}$$

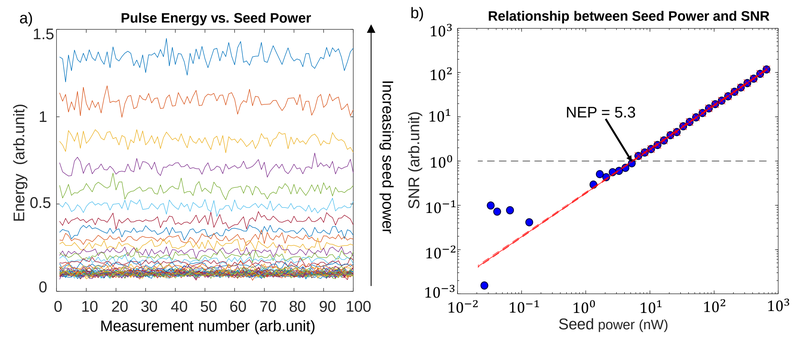


Figure 5 (a): Energy in each pulse over 100 measurements for a variety of seed powers. (b) SNR versus seed power. NEP indicated by arrow is 5.3 nW for this example.

Results

Summary Figure

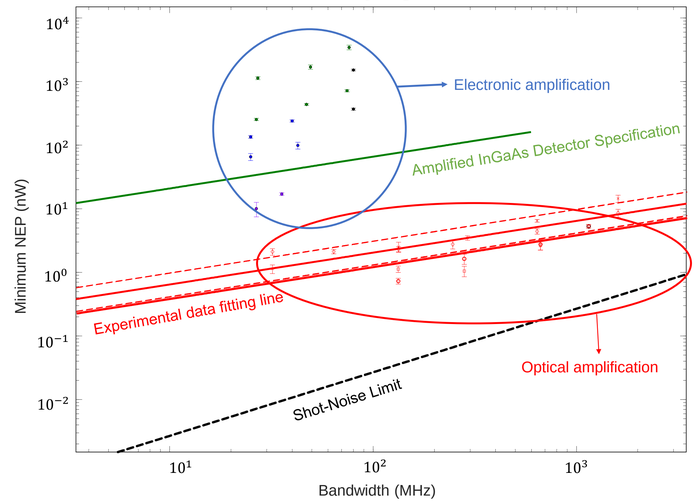


Figure 6: Summary of minimum NEP as a function of integration bandwidth. Optical amplification data (red) include measurements for 50 m (dashed fitting lines) and 480 m (solid fitting lines). Control experiments (using PD and different transimpedance amplifiers) shown within blue circle. Comparison metric calculated for a commercial Amplified InGaAs detector [5] shown in green. Quantum noise limit shown by black dashed line.

Minimum NEP (in $fW/\sqrt{HZ}$)		
Clock Speed	Fiber Length	
	50 m	480 m
3.2GS/s	304	204
12.8GS/s	129	125
Amplified PD [5]	6600	

Table 3: Minimum NEP in experiments compared to a commercial amplified photo diode [5].

Conclusion

Performance

- 10X bandwidth and sensitivity relative to state-of-the-art detectors.
- Resolve ~20 photons at 1.6 GHz bandwidth.
- Drop-in compatible for any light detection system.
- Low pump power required (~2 W) → CW pumping possible.

Outlook

- Reduction in pump noise to further increase sensitivity.
- Extend to other wavelengths, bandwidths.
- Migrate to an on-chip platform with silicon photonics.
- Photon-starved communications, sensing, and bio-imaging.

References

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